

From Scalar Reward to Cellular Credit: Separating Representation Selection from Neuron-Specific Learning

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Abstract

A delayed reward can show that an action was useful, but it does not say how each neuron should change. We examined this problem in abstract closed-loop brain–computer interface simulations motivated by the role-dependent dendritic signals reported by Francioni et al. [Francioni et al., 2026]. The studies asked two related questions. Can scalar reward select and retain a useful population activity pattern? Can it also learn a different sign of change for every cell when those signs are not known to the learner? The pART-inspired models answered the first question. They selected useful patterns from a repertoire, maintained them across a delay, and relearned them after the hidden cell roles changed. However, the complete theory-led model failed its main preregistered test. Its early dendritic signal did not predict later synaptic and somatic changes. We also found that its learned neuron-by-neuron pattern was inherited almost exactly from activity patterns already present in the repertoire. A separate comparator tested a generic local-eligibility rule that is not part of the cited Grossberg theories. Every cell received the same scalar reward signal, but each cell combined it with a private record of its own recent perturbation. This rule learned arbitrary hidden sign maps and relearned them after remapping. At population size $N = 32$, alignment was 0.987 after acquisition and 0.964 after remapping. Scalar reward was therefore sufficient for representation selection in the tested pART-inspired models. It was sufficient for arbitrary cellular credit only when combined with a cell-specific eligibility trace. The simulations do not establish that biology uses this rule.

1 Introduction

Neural learning faces two related credit-assignment problems. First, an outcome may arrive after the action that caused it. The learner must connect an earlier event with that later outcome. Second, many neurons may be active at the same time, even though they contribute differently to the outcome. The learner must determine how each one should change.

We call selection at the population level *representation credit*. It means choosing and reinforcing a useful activity pattern that is already available to the system. We use *cellular credit* for the finer problem of deciding the sign and size of change at each cell. The arrangement of those cell-specific effects is called the *topology* of the learned signal. A system can solve representation credit without constructing a new cellular topology.

This distinction is especially important in brain–computer interface (BCI) tasks in which nearby cortical cells have opposite causal roles. Francioni et al. reported role-dependent dendritic signals during such learning and interpreted them as vectorized instructive signals [Francioni et al., 2026].

One possible mechanism would compute a separate teaching value for every neuron. Another would broadcast one scalar outcome and let information stored locally at each cell determine its update.

pART, ART, and SMART include components that make the second possibility worth testing. They provide mechanisms for category selection, maintenance across delay, learned expectations, matched feedback, and local plasticity. None of those components alone explains how an arbitrary cell-specific sign pattern can originate when the environment returns only a scalar result.

We addressed that question through a sequence of computational studies. Each study addressed a limitation of the preceding one. We first asked whether selective feedback could route an opponent pattern that was already built into the model. We then hid cellular identity and required the model to select from random population patterns. Next, the model learned its own top-down expectation instead of receiving a copied pattern. A reduced local plasticity mechanism was tested on its own before it was added to the complete model. We then traced the source of the neuron-by-neuron pattern learned by that model. The final study tested whether a local record of recent exploration could supply the missing cell-specific information.

This order is important. Successful behavior alone does not reveal whether the learner selected an existing representation or discovered how individual cells should change.

The contribution is therefore both computational and methodological. The experiments separate representation selection from cellular sign learning, evaluate a prospective dendrite-to-synapse-to-soma criterion, and identify the additional information that made the generic cellular solution possible. All numerical results and figures come from the archived repository record. Claims about published theory and biology are supported separately by external citations.

2 Related work

ART addresses the stability-plasticity problem through category matching, vigilance, reset/search, and learned top-down expectations [Carpenter and Grossberg, 1987]. pART emphasizes the structural and temporal aspects of credit assignment. A representation can be selected, maintained over a delay, and reinforced by broad motivational signals [Grossberg, 2018]. These mechanisms motivate representation-level credit but do not, by themselves, specify an arbitrary neuron-resolved feedback vector.

SMART provides a more cellular circuit motif. In a visual thalamocortical setting, learned higher-to-lower feedback reaches apical dendrites, a match-dependent on-center/off-surround circuit changes timing and participation, and a local voltage/conductance-gated rule changes synapses [Grossberg and Versace, 2008]. Canonical laminar extensions describe related modulatory feedback motifs [Grossberg, 2021]. Grossberg’s vector associative motor systems address a different information boundary. They use an available target-versus-present difference vector [Gaudiano and Grossberg, 1991]. Importing such a vector into the present task would assume the neuron-specific information that the simulations are intended to explain.

Three computational accounts must consequently be distinguished. A centralized system may compute and route a cell-wise error. A representation-level system may reinforce one useful distributed pattern already present in a repertoire. A local exploratory system may instead multiply a common scalar advantage by a cell-specific eligibility trace. The experiments below test the latter two possibilities without treating them as equivalent.

3 Methods

3.1 Task and available information

The experiments used abstract closed-loop BCI tasks. Each cell had a hidden causal role $c_i \in \{-1, +1\}$, and the two roles occurred equally often. The learner could observe the current context, its own activity, permitted local variables, and a scalar outcome. The learner could not observe the hidden role vector c . That information was used only in labeled positive controls and offline analyses. We measured task performance, alignment between learned patterns and the hidden roles, recovery after the roles were changed independently, and the effects of removing proposed causal mechanisms. An alignment of 1 indicates a perfect match with the hidden roles. We use *remapping* for an independent change in those roles.

The theory-led models contain five quantities. The selected representation is $H_{k,h}$. Its learned top-down expectation is $T_{k,h,i}$. The modeled dendritic input is D_i , the plastic lower weight is W_i , and the somatic output is S_i . The main preregistered test asked whether an early dendritic signal predicted a later weight change and whether that weight change predicted a later change in somatic output. The predicted sequence was

$$D_{\text{residual},i}^{\text{early}} \longrightarrow \Delta W_i \longrightarrow \Delta S_i. \quad (1)$$

The test was evaluated while holding the selected hypothesis fixed. This prevents a switch between population patterns from appearing to be a change within individual cells. The dendritic measure was a *residual*, meaning the part of the modeled dendritic signal left after accounting for somatic activity.

The SMART-derived models used the following qualitative plasticity rule.

$$\Delta W_{jk} \propto f_G(V_k, \bar{g}_{jk}) [\bar{g}_{jk} f_N(V_k)(\hat{w} - \check{w}) + w_0 - w_{jk}]. \quad (2)$$

In words, a synaptic weight changes according to conductance, voltage, spike timing, and the weight’s current value [Grossberg and Versace, 2008]. These are quantities available at or near the modeled synapse.

The generic local-eligibility comparator used a different update.

$$\Delta A_i = \eta_N (R - b) e_i. \quad (3)$$

Here, $R - b$ is the same scalar reward advantage for every cell. The eligibility trace e_i is a short-lived record of cell i ’s own exploratory perturbation. Because this trace differs across cells, the same reward can produce different local updates. The factor η_N corrects known population-mean scaling. Hidden roles do not enter the update.

3.2 Experimental sequence and reproducibility

Later studies used frozen held-out seed sets, preregistered effect floors, seed-level bootstrap intervals, explicit positive controls supplied with hidden vector information, and causal ablations that shuffled outcomes, removed exploration, and removed eligibility. Table 1 provides the only manuscript-level crosswalk to the repository’s internal identifiers. The Results retain this chronology but use scientific study names in the narrative.

Registry ID	Study	Purpose
EXP000	Prestructured opponent-code routing	Existence proof for routing an already available pattern.
EXP001	Random-repertoire selection	Scalar selection and delayed credit at the representation level.
EXP002	Learned top-down expectancy	Test of whether a learned expectation supplies the longitudinal cellular signal.
EXP003– EXP003b	SMART motif and integrated composition	Test local plasticity, then test the complete model.
EXP004	Origin of learned topology	Test of construction versus inheritance from fixed repertoire geometry.
EXP005	Published-source comparison and local eligibility	Check the published mechanisms, then test generic cellular credit.

Table 1: Crosswalk between repository identifiers and the scientific studies described in the manuscript.

3.3 Scope of the claims

Quantitative statements labeled as results are derived from committed protocols, frozen analyses, held-out records, or explicitly identified post-hoc audits in the project repository [OgnjenX, 2026]. No new simulations were conducted specifically for this manuscript. The prestructured study tests routing, not de novo sign learning. Conclusions from theory-led studies apply to the implemented compositions and endpoints rather than to all ART, pART, or SMART models. The local-eligibility comparator is a generic three-factor rule and is not attributed to Grossberg. Biological interpretations remain hypotheses unless supported by the cited experimental literature.

3.4 Link to published theory

We compared every implemented step with the cited sources. pART motivated representation selection and maintenance. ART motivated expectation learning and match/reset. SMART motivated apical feedback and timing-dependent local plasticity. The dendritic measure was motivated by Francioni et al. The sources did not explain how a scalar outcome alone could create an arbitrary neuron-by-neuron mapping from H to T . They also did not establish two biological mappings used by the model. The first links the SMART surround to NDNF-sensitive dendritic signals. The second links modeled current or voltage to a soma-conditioned calcium residual. These gaps motivated the later tests of topology and local eligibility.

4 Results

4.1 Routing a pattern already present

The first study corresponds to registry record EXP000. It asked whether selective feedback could express opposite effects in two groups of cells when the required pattern was already built into the model. After two retained calibration attempts, the corrected model reached a late-task accuracy of 0.978 across 30 seeds. P+ modulation was +0.673, and P- modulation was −0.673 (Figure 1). Their difference, the opposition index, was +1.346. Selective feedback could therefore route a strong opponent pattern without receiving a separate error value for every neuron on each trial.

This was an existence proof, not a test of cell-specific learning. P+/P- identity could be recovered from even and odd neuron indices, and that same identity was encoded in the category templates. Freezing category and value learning did not reduce accuracy. Every ordinary corrected trial matched a stored category, and none triggered a reset (Table 2). The model could use a pattern that it already possessed. It did not explain how that pattern could be found when cellular identity was hidden.

4.2 Selecting from a random repertoire

The next study (EXP001) hid population identity and replaced the opponent template with a bank of random patterns. It asked whether a scalar outcome could select a useful pattern, preserve it across irrelevant events, and support selection of another useful pattern after the hidden roles changed. The model succeeded at all three tasks.

Success depended strongly on what the random bank contained. Larger banks were more likely to include a pattern that already aligned well with the hidden roles, and the best initial alignment increased with bank capacity (Figure 2). This result places the achievement at the representation level. Reinforcement selected a useful direction, but that direction had to be present before learning began.

The dendritic residual did not resolve the remaining cellular-credit problem. Its instantaneous alignment with hidden roles also appeared in controls that did not learn, and it did not reliably predict later cell-specific activity change. A positive control supplied with a separate teaching value for every neuron did predict that change. The next study asked whether a learned top-down expectation could generate the missing signal without receiving the hidden role vector.

4.3 Learning a top-down expectation

The learned-expectancy study (EXP002) asked whether repeated experience with selected patterns could produce a useful top-down signal. It added a Grossberg outstar-derived association,

$$T_h \leftarrow T_h + \eta(S_{\sim} - T_h), \quad (4)$$

This equation moves the expectation T_h toward the somatic pattern $S_{\sim}$ repeatedly active with the selected hypothesis. The expectation was learned and could be relearned after remapping. The behavioral scores and top-down alignments are reported in Table 2.

These behavioral results did not establish that the learned expectation carried a cellular teaching signal. Directly copying the selected pattern produced similar alignment, and a contextual bandit matched the behavioral performance. The main within-cell test also failed. Before and after remapping, the dendritic residual effects were -0.326 and -0.016 . The corresponding effects in the neuron-indexed positive control were 0.660 and 0.545 . Suppressing expectation learning removed the alignment of the learned pattern but had little effect on behavioral performance.

The model could store an expectation, but the expectation did not yet explain later cell-specific change. The next step was to test whether matched feedback could control local plasticity.

4.4 Testing local plasticity and the complete model

The local-mechanism study began with a comparison to the published theory (EXP003) and then tested a reduced SMART-inspired motif (EXP003a). It asked a narrow question. Could supplied matched feedback alter spike timing, permit a local weight update, and change a later feedforward response? The model did all three. The update and spike-probe results are reported in Table 2.

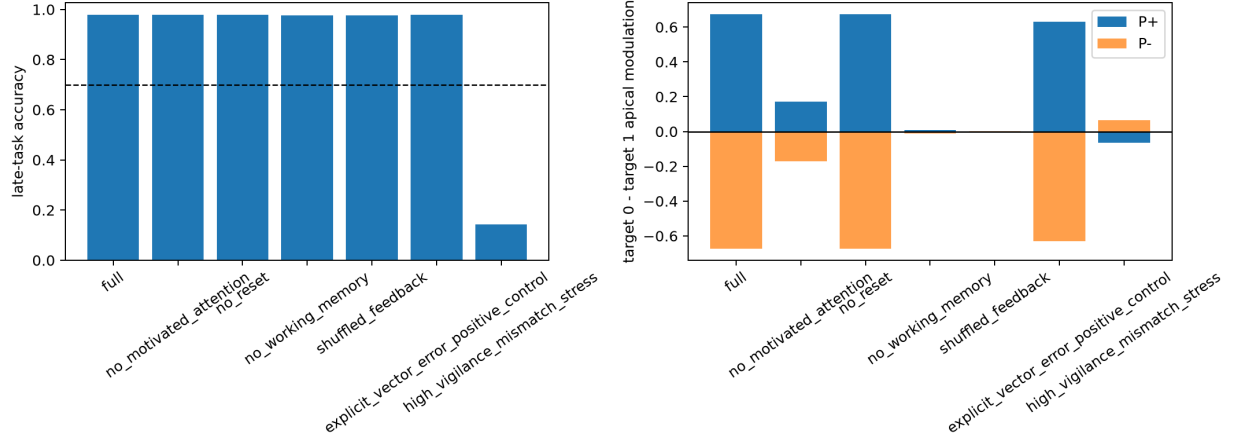


Figure 1: Repository-derived summary of prestructured routing. Late-task accuracy remains high under several ablations because the opponent pattern is already present in the model. Shuffled feedback disrupts P+/P- modulation, while the explicit vector control serves as a positive control.

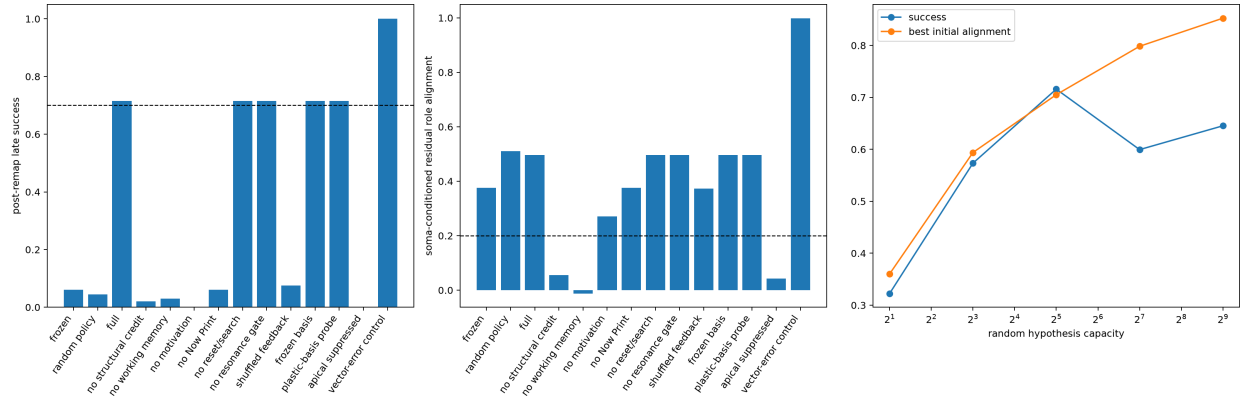


Figure 2: Frozen random-repertoire comparison. The left panel shows post-remap success by condition, and the center panel shows soma-conditioned residual alignment. The right panel relates success and the best initial role alignment to hypothesis-bank capacity. The capacity relation is consistent with selection from a fixed repertoire rather than de novo cellular sign discovery.

No update occurred under competitor, top-down-ablated, or mismatch conditions. This result established the proposed local sequence when the appropriate feedback signal was supplied. It did not explain how the system learned that signal.

The complete model (EXP003b) combined this local mechanism with delayed structural credit and learned expectations. Across 12 held-out mappings, feedback advanced local spike timing and changed lower weights. Removing SMART input, the learned expectation, or local learning eliminated the effect. The timing result is reported in Table 2.

However, the model failed its main preregistered test. The early dendritic residual did not predict the later weight change or the later somatic change well enough to meet the predefined threshold. The neuron-wise positive control did complete this predictive chain. A post-hoc analysis then asked where the prediction was lost. The learned expectation T and variables close to the plasticity rule predicted the sparse weight changes, but the soma-conditioned residual did not. Raw $T \rightarrow \Delta W$ correlation was 0.416 [0.290, 0.530]. Residual $D \rightarrow \Delta W$ correlation was -0.004 [-0.141 , 0.128]. This analysis did not change the failed classification. Instead, it pointed to two separate questions. Where did the expectation’s topology come from, and did the residual measure preserve the relevant information?

4.5 Tracing the learned neuron pattern

The topology study (EXP004) asked whether the complete model constructed a neuron-by-neuron pattern or instead inherited one from its fixed repertoire. The test varied the quality of the best pattern available before learning. Better initial coverage produced better behavior and better alignment between the learned expectation and the hidden roles (Figure 3). At bank size $M = 16$, the behavioral difference between high and low coverage was 0.197 [0.137, 0.255]. The corresponding scores and alignments are reported in Table 2.

A constructed sequential condition showed that the model could combine behavior across steps. Its performance reached 0.980 and exceeded the best repeated-single action by 0.313 [0.260, 0.374]. However, a contextual bandit reached 0.977. Behavioral composition alone therefore did not show that the model had constructed a new cellular topology.

The source of the learned pattern became clear through a comparison with experienced activity.

$$\text{corr}(T, \text{selected initial soma}) = 0.999851, \quad \text{corr}(T, \text{mean target}) = 0.999870. \quad (5)$$

Exact replay reconstructed T with RMSE 0.0, and T did not surpass the best individual target that the model had experienced. Permuting outcomes changed which hypotheses and targets were visited, accounting for most of the change in topology. The outstar learned from selected states, but the neuron-by-neuron coordinates of those states came mainly from the fixed repertoire. Learning an arbitrary cellular topology would require a cell-varying quantity that was not already present in a candidate pattern.

4.6 Testing a local eligibility trace

The final study (EXP005) first asked whether a published Grossberg equation supplied the missing cell-varying quantity. pART, ART/outstar learning, exploration, SMART, and vector motor systems each provided a related component. None of the inspected sources combined an independent local exploratory trace with a delayed scalar outcome to learn arbitrary intermingled signs. We therefore did not construct another model and label it as a Grossberg model.

Instead, a separate generic comparator tested whether that additional operation was computationally sufficient. After each exploratory change, each cell retained a private eligibility trace e_i .

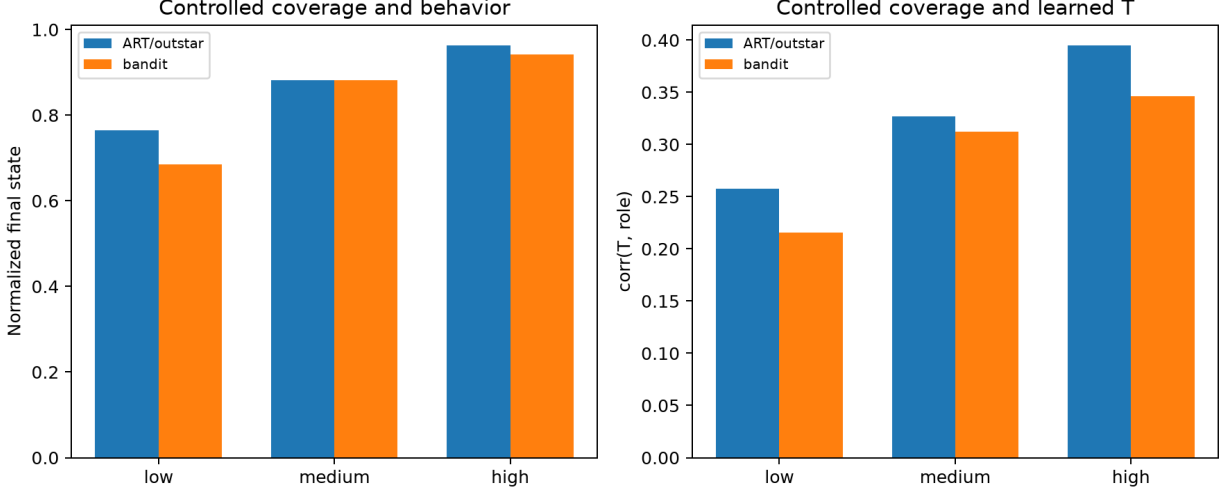


Figure 3: Frozen controlled-coverage result. Both final behavior (left) and alignment of the learned top-down expectation T with hidden roles (right) increase with the quality of topology already available in the bank. The contextual-bandit comparison follows the same trend.

A later scalar outcome strengthened changes that had helped and reversed changes that had hurt. No hidden role information entered the update.

The comparator learned the hidden signs across all four tested population sizes and relearned them after an independent remap (Table 3). At $N = 32$, alignment reached 0.987 after acquisition and 0.964 after remapping. The learning curve rose quickly, fell when the hidden map changed, and rose again as the new map was learned (Figure 4). Shuffling the outcome reduced post-remap alignment to 0.229. Removing plasticity, the temporal eligibility trace, or exploration left alignment near zero. The difference between the generic and shuffled conditions was 0.735 [0.655, 0.820].

Exact replay reproduced the updates to numerical precision. Final topology also exceeded the best single exploratory sample at every population size. The comparator therefore accumulated information across trials rather than selecting one fortunate perturbation. The scalar reward carried no more information than before. The new information came from the local trace, which allowed the same reward to produce different updates at different cells.

Study	Supporting comparison	Result
Routing	Accuracy with category and value learning frozen	0.978
Routing	Stored-category matches on ordinary corrected trials	100%, no resets
Learned expectation	Behavior before and after remapping	0.528, 0.397
Learned expectation	Top-down role alignment before and after remapping	0.343, 0.287
Local plasticity	Modeled update and learning-off spike probe	$0.600 \rightarrow 0.731$ and $0 \rightarrow 7/8$
Complete model	Feedback-induced advance in local spike timing	6.29 ms
Topology	Behavior at low, medium, and high initial coverage	0.765, 0.882, 0.962
Topology	Learned- T alignment at the same coverage levels	0.258, 0.327, 0.395

Table 2: Supporting quantitative results from the theory-led studies.

N	Acquisition alignment	Post-remap alignment	Changed-cell sign reversal
8	0.997	0.968	1.000
16	0.993	0.960	1.000
32	0.987	0.964	1.000
64	0.972	0.950	0.975

Table 3: Generic local-eligibility results across population sizes. Each value summarizes 16 held-out maps after 1,280 acquisition episodes and 2,560 remap episodes.

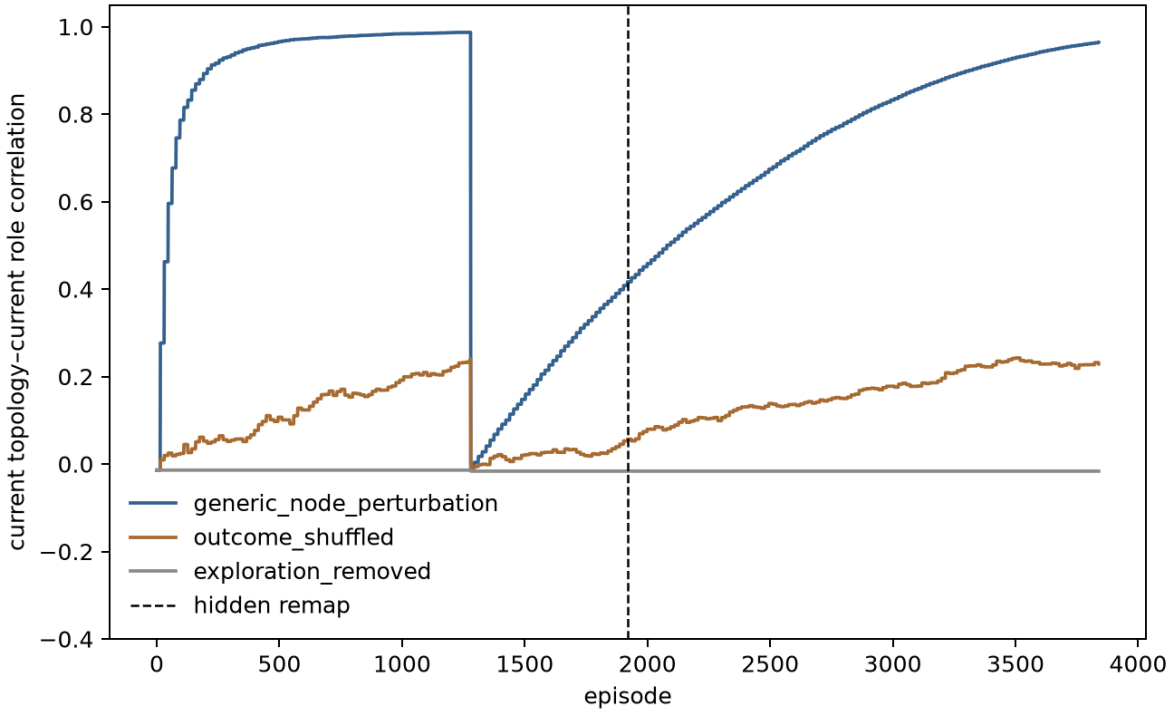


Figure 4: Frozen generic local-eligibility diagnostic at $N = 32$. The role alignment learned from scalar outcome and local perturbation eligibility rises during acquisition, drops when the hidden map changes, and is relearned. Shuffled outcome learns only weak alignment, while removing exploration eliminates learning.

5 Discussion

The experiments separate two kinds of learning that can produce similar behavior. Scalar reinforcement was enough to select and maintain useful population patterns. It also supported learned expectations and remapping. However, the neuron-by-neuron structure expressed by these models came mainly from patterns that were already available for selection. Selecting a successful representation is not the same as learning how each cell should change.

The generic comparator showed how one scalar outcome can support different updates at different cells. Every cell received the same reward signal, but each combined it with a private memory of its own recent perturbation. Over repeated trials, these local associations formed the hidden neuron-by-neuron pattern. A centrally routed vector error is therefore not mathematically necessary in this abstract task. This result does not show which mechanism the brain uses. Biological learning could depend on local eligibility, explicit cell-specific instruction, matched expectations, or some combination of them.

The complete model also revealed a measurement problem. Information can be present in a learned expectation, a current, or a spike-timing variable and still be absent from the measured calcium residual. The later analysis helped locate where the information was lost. It could not turn the failed preregistered test into a positive result. Future studies should state in advance which variable is expected to cause learning and how that variable is expected to appear in the measured dendritic signal.

The comparison with published theory also sets an important boundary. A model may resemble pART or SMART without every component following from those theories. The cited work motivates representation selection, learned expectations, matching, and local plasticity. The exploratory eligibility rule adds a new hypothesis. It should be tested as a separate mechanism rather than attributed to Grossberg.

6 Limitations

These tasks are abstract simulations. They do not reconstruct retrosplenial circuitry, animal behavior, or the complete equations of pART and SMART. Several biological links remain assumptions. These include connecting a pART-selected representation to a SMART-like projection, targeting the relevant BCI cells, mapping center/surround organization onto NDNF circuitry, and converting modeled activity into a calcium residual. The first routing model also begins with a favorable structure. The random-repertoire and outstar results depend on the particular pattern banks and ownership rules that were tested.

The complete model used sparse, quantized local plasticity. Later analyses help explain why its main test failed, but they do not change the original classification. The local-eligibility comparator also uses a known correction for population size. That correction contains no hidden role information. Even so, success in a finite, normalized task does not show that the rule would scale biologically. Finally, this manuscript synthesizes archived computational records. It is not an independent reproduction or peer review of either the simulations or the motivating animal experiment.

7 Biological predictions and discriminating tests

The competing explanations make different experimental predictions.

1. **Interaction between eligibility and outcome.** A cell-specific deviation before the outcome should leave a local trace. The interaction between that trace and the later scalar

outcome should predict the sign of plasticity. Disrupting the interval between the deviation and the outcome should impair learning without eliminating the scalar reward response.

2. **Remapping within the same cell.** If selected feedback determines the expressed pattern, a change in context or reward contingency should reorganize feedback before later somatic activity changes. The same cell should also be able to reverse its role across contexts.
3. **Measurement of center and surround.** Excitatory feedback and NDNF-associated inhibitory surround should be measured and perturbed independently. This would test whether the predictive signal is a signed membrane effect rather than an excitatory calcium signal alone.
4. **Intervention at a specific projection and time.** A candidate feedback source should be perturbed before the outcome while sensory error and reward are preserved. This would show whether that projection carries cell-specific information. Broad suppression of activity would be harder to interpret.

Together, these tests could distinguish among cell-specific instruction, selection of an available population pattern, and credit computed from a local exploratory trace.

8 Conclusion

Scalar outcome was enough to select and maintain useful population patterns without a separate teaching value for every neuron. In the tested pART-inspired models, however, the neuron-by-neuron pattern came mainly from representations that were already available. The complete model also failed its preregistered test linking an early dendritic signal to later synaptic and somatic change. A separate generic rule learned an arbitrary cellular pattern by combining scalar outcome with a local eligibility trace. The central result is therefore the distinction between selecting a representation and assigning credit to individual cells. The latter problem requires some source of cell-varying information. Whether dendrites store that information locally, receive it as an explicit vector, or express it through matched feedback remains an empirical question.

Data and code availability

All computational results reported here derive from archived protocols, frozen analyses, held-out experiment records, and explicitly labeled post-hoc audits associated with this study. No new simulation results were generated specifically for the manuscript. The public project repository provides the source code, experimental records, analysis outputs, and figure files used in this manuscript.

References

- Gail A. Carpenter and Stephen Grossberg. ART 2: Self-organization of stable category recognition codes for analog input patterns. *Applied Optics*, 26(23):4919–4930, 1987. doi: 10.1364/AO.26.004919.
- V. Francioni et al. Vectorized instructive signals in cortical dendrites during a brain–computer interface task. *Nature*, 652:1254–1263, 2026. doi: 10.1038/s41586-026-10190-7.
- Paolo Gaudiano and Stephen Grossberg. Vector associative maps: Unsupervised real-time error-based learning and control of movement trajectories. *Neural Networks*, 4(1):147–183, 1991. doi: 10.1016/0893-6080(91)90002-M.
- Stephen Grossberg. Desirability, availability, credit assignment, category learning, and attention. *Brain and Neuroscience Advances*, 2, 2018. doi: 10.1177/2398212818772179.
- Stephen Grossberg. A canonical laminar neocortical circuit whose bottom-up, horizontal, and top-down pathways control attention, learning, and prediction. *Frontiers in Systems Neuroscience*, 15:650263, 2021. doi: 10.3389/fnsys.2021.650263.
- Stephen Grossberg and Massimiliano Versace. Spikes, synchrony, and attentive learning by laminar thalamo-cortical circuits. *Brain Research*, 1218:278–312, 2008. doi: 10.1016/j.brainres.2008.04.024.
- OgnjenX. part-vector-credit: part-inspired credit assignment versus dendritic vectorization. GitHub repository, 2026. URL <https://github.com/OgnjenX/part-vector-credit>. Repository experimental record inspected 15 September 2026.